STAFF REVIEW · H446 052 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. An existing node reference.

2. Its first node.

3. Left subtree keys less and right keys greater under chosen policy.

4. 12 is the left child of 15. Its search path is right from 10, left from 18, then left from 15.

I Do — source worked steps

1. Explain the trace — 7 becomes right child of 6; follow comparisons until a null child is reached.

We Do — guided CHECK

1. Explain the trace
Place 12 as left child of 14. Tree ordering remains valid.

You Do — source mark schemes

1. 6<8 [2]; 6>3 [2]; right child of3 [2].

2. 3,6,8,10 [4]; left-root-right order follows BST invariant [2].

3. Balanced height ≈log n [2]; follow only one path per comparison [1]; sorted insertions may create chain height n [2]; search may inspect n nodes [1].

Exit Ticket — source answer

1. False. It can become a chain.